

■ 1.2.2 Defining Variables

When you do long calculations, it is often convenient to give *names* to your intermediate results. Just as in standard mathematics, or in other computer programming languages, you can do this by introducing named *variables*.

This sets the value of the <i>variable</i> <code>x</code> to be 5.	<code>In[1]:= x = 5</code> <code>Out[1]= 5</code>
Whenever <code>x</code> appears, <i>Mathematica</i> now replaces it with the value 5.	<code>In[2]:= x ^ 2</code> <code>Out[2]= 25</code>
This assigns a new value to <code>x</code> .	<code>In[3]:= x = 7 + 4</code> <code>Out[3]= 11</code>
<code>pi</code> is set to be the numerical value of π to 40 digit accuracy.	<code>In[4]:= pi = N[Pi, 40]</code> <code>Out[4]= 3.141592653589793238462643383279502884197</code>
Here is the value you defined for <code>pi</code> .	<code>In[5]:= pi</code> <code>Out[5]= 3.141592653589793238462643383279502884197</code>
This gives the numerical value of π^2 , to the same accuracy as <code>pi</code> .	<code>In[6]:= %^2</code> <code>Out[6]= 9.8696044010893586188344909998761511353137</code>

<code>x = value</code> assign a value to the variable <code>x</code> <code>x = y = value</code> assign a value to both <code>x</code> and <code>y</code> <code>x = .</code> remove any value assigned to <code>x</code>

Assigning values to variables.

It is very important to realize that values you assign to variables are *permanent*. Once you have assigned a value to a particular variable, the value will be kept until you explicitly remove it. The value will, of course, disappear if you start a whole new *Mathematica* session.

Forgetting about definitions you made earlier is one of the most common sources of confusion when using *Mathematica*. If you set `x = 5`, *Mathematica* assumes that you *always* want `x` to have the value 5, until or unless you explicitly tell it otherwise. Particularly when you are using variables to represent symbolic quantities

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(see Section 1.4), it is very important that you remove any values you have defined when you finish using them.

Values you assign to variables stay until you explicitly change or remove them.

An important point to remember when using *Mathematica*.

The variables you define can have almost any names. There is no limit on the length of their names. One constraint, however, is that variable names can never *start* with numbers. `x2` could be a variable, but `2x` means `2*x`.

Mathematica uses both upper and lower-case letters. You should make sure, however, that the names of your variables at least begin with lower-case letters. This will avoid confusion with built-in *Mathematica* objects, such as those in Section 1.1.3, whose names always start with upper-case letters.

<code>aaaaa</code>	a variable name containing only lower-case letters
<code>Aaaaa</code>	a built-in object whose name begins with a capital letter

Naming conventions.

You can type formulae involving variables in *Mathematica* almost exactly as you would in mathematics. There are a few important points to watch, however.

`x y` means `x` times `y`
`xy` with no space is the variable with name `xy`
`5x` means 5 times `x`
`x^2y` means $(x^2) y$, not $x^{(2y)}$

Some points to watch when using variables in *Mathematica*.